

Twelfth Section. The bitangents of a curve of the fourth order.

§ 95. Number of bitangents of a curve of the fourth order.

A geometrical problem which, because of its many relations with other domains, is of special importance, and which at the same time, in a manner similar to the problem of the inflection points of curves of the third order, offers remarkable algebraic relations, is the problem of the bitangents of curves of the fourth order.

By a *bitangent* of a curve one understands, as was already observed in § 86, a straight line which touches the curve at two different points. For curves of order lower than the fourth, bitangents cannot occur. For curves of the fourth order a bitangent has, apart from the points of contact, no point in common with the curve. In special cases the two points of contact may also coincide. Then we have lines with four-point contact.

As an algebraic problem the determination of the bitangents depends on an algebraic equation whose degree is equal to the number of bitangents; the first question is therefore the degree of this equation, that is, the number of bitangents. We restrict ourselves here to curves of the fourth order, although the method which we use is also applicable to curves of higher order. We also assume that the curve of the fourth order is free from singular points.¹

Let

$$f(x_1, x_2, x_3) = 0 \quad (1)$$

or, more briefly, $f(x) = 0$, be a ternary form of the fourth degree which, when equated to zero, represents a curve of the fourth order without singular point; we shall call it the curve f . If, in this equation, one puts in place of x_1, x_2, x_3 expressions of the form

$$x_1 + ty_1, \quad x_2 + ty_2, \quad x_3 + ty_3,$$

there results an equation of the fourth degree with respect to t ,

$$f(x + ty) = 0, \quad (2)$$

whose roots, when (x) and (y) are two fixed points, give, when inserted in $x + ty$, the coordinates of the intersections of the joining line of the points $(x), (y)$ with the curve f . We shall denote this line by (x, y) .

Now order the function $f(x + ty)$ according to powers of t . This gives, as in Volume I, § 60,

$$f(x + ty) = P_0(x, y) + 4tP_1(x, y) + 6t^2P_2(x, y) + 4t^3P_3(x, y) + t^4P_4(x, y), \quad (3)$$

where the $P_\nu(x, y)$ are the polars of $f(x)$, hence homogeneous functions of degree ν in y and of degree $4 - \nu$ in x . Thus

$$\begin{aligned} P_0(x, y) &= f(x), \\ P_1(x, y) &= \frac{1}{4} \sum_i y_i \frac{\partial f}{\partial x_i}(x), \\ P_2(x, y) &= \frac{1}{4 \cdot 3} \sum_{i,k} y_i y_k \frac{\partial^2 f}{\partial x_i \partial x_k}(x), \\ P_3(x, y) &= \frac{1}{4 \cdot 3 \cdot 2} \sum_{i,k,l} y_i y_k y_l \frac{\partial^3 f}{\partial x_i \partial x_k \partial x_l}(x), \\ P_4(x, y) &= f(y). \end{aligned} \quad (4)$$

¹Jacobi, Clebsch, Hesse, Steiner, Aronhold, Geiser, Cayley, Salmon, Riemann, Weber, Frobenius, and Noether are the principal references named in the source passage for the theory of bitangents of a plane quartic.

If x lies on the curve f , then $P_0 = 0$, and one root of (2) vanishes. If, in addition, (y) is taken on the tangent at (x) , then also $P_1(x, y) = 0$, and two roots of (2) vanish. The other two are determined by

$$6P_2 + 4tP_3 + t^2P_4 = 0. \quad (5)$$

If this quadratic equation has two equal roots, then the line (x, y) will have a second point of contact with the curve f , that is, it will be a bitangent. The condition for this is that the discriminant of (5) vanish, or that

$$R(x, y) = 2P_3(x, y)^2 - 3P_2(x, y)P_4(x, y) \quad (6)$$

be equal to zero. The equation $R = 0$ is also satisfied when (x) is an arbitrary point of the curve and (y) coincides with (x) ; otherwise R vanishes only when the line (x, y) is a bitangent. Hence:

The equation $R(x, y) = 0$, when (x) is a point of the curve f and (y) is a point of the tangent at (x) different from (x) , is the necessary and sufficient condition that (x) be a point of contact of a bitangent.

It remains to derive from this condition another one containing only the x 's, and satisfied by no points of the curve other than the points of contact of the bitangents. The variables y satisfy the tangent equation

$$y_1f'_1(x) + y_2f'_2(x) + y_3f'_3(x) = 0. \quad (7)$$

Let b_1, b_2, b_3 be arbitrary constants and put

$$b_x = b_1x_1 + b_2x_2 + b_3x_3. \quad (8)$$

Together with (7) we add the equation $b_1y_1 + b_2y_2 + b_3y_3 = 0$. Thus (y) is the intersection of the tangent at (x) with the arbitrary line $b_x = 0$, and we obtain

$$y_1 = b_2f'_3 - b_3f'_2, \quad y_2 = b_3f'_1 - b_1f'_3, \quad y_3 = b_1f'_2 - b_2f'_1. \quad (10)$$

Through this substitution $R(x, y)$ becomes a homogeneous function $D(x, b)$ of degree twenty in x and degree six in b . Besides the points of contact of the bitangents, it also vanishes at the four intersections of the line b_x with the curve f .

If on the tangent we replace y by $\mu x + \lambda y$, then, for $f(x) = 0$, one has

$$R(x, \mu x + \lambda y) = \lambda^6 R(x, y). \quad (11)$$

The point $(\mu x + \lambda y)$ may be any point of the tangent. Repeating the construction with another arbitrary line

$$a_x = a_1x_1 + a_2x_2 + a_3x_3 \quad (13)$$

gives

$$\frac{D(x, a)}{a_x^6} = \frac{D(x, b)}{b_x^6}. \quad (14)$$

Since this is valid only on $f(x) = 0$, it yields the identity

$$b_x^6 D(x, a) - a_x^6 D(x, b) = f(x) Q(x, a, b). \quad (15)$$

A comparison in a coordinate system in which a_x and b_x are coordinate axes shows that

$$Q(x, a, b) = a_x^6 Q_1(x) - b_x^6 Q_2(x).$$

Consequently there is a form $\chi(x, a, b)$ of the fourteenth degree such that

$$\frac{D(x, a) - f(x) Q_1(x)}{a_x^6} = \chi(x, a, b). \quad (16)$$

After subtracting the relation obtained by giving a, b fixed values, one obtains a form $X(x)$, rationally dependent on the coefficients of f , for which

$$D(x, a) = a_x^6 X(x) + f(x) \Phi(x, a). \quad (18)$$

Thus the curve

$$X(x) = 0 \quad (19)$$

of the fourteenth degree passes through the contact points of the bitangents and through no other points of $f = 0$. It follows that:

A curve of the fourth order without singular point has twenty-eight bitangents.

Indeed, the intersections of $X = 0$ with $f = 0$ amount to $14 \cdot 4 = 56$ points, and the two contact points of each bitangent are counted. The only remaining doubt would be that the curve X could touch f , or that f could pass through singular points of X . The following considerations remove this doubt by displaying forms of the equation of the curve for which the existence of twenty-eight distinct bitangents is manifest.

§ 96. The Steiner complexes.

If $x_1 = 0$ is the equation of a bitangent of the curve of the fourth order $f = 0$, then, after putting $x_1 = 0$, f must become a square. Hence f must have the form

$$f = x_1 V - u^2, \quad (1)$$

where V is cubic and u is quadratic. But the function f can be put into the form (1) in a triply infinite number of ways. If p is any linear function, then

$$f = x_1 (V + 2pu + p^2 x_1) - (u + px_1)^2,$$

again has the form (1).

Let $y_1 = 0$ be a second bitangent distinct from x_1 . The constants in p may be chosen so that the conic $u + px_1 = 0$ passes through the two points of contact of y_1 . Then the cubic form V must vanish at these same points. Since $y_1 = 0$ is a bitangent, the cubic $V = 0$ is touched there by the line $y_1 = 0$; hence V must split and contain y_1 as a factor. Therefore

$$f = x_1 y_1 v - u^2, \quad (2)$$

where u, v are quadratic forms. Conversely, if x_1, y_1 are any two bitangents of a quartic $f = 0$, then f can be brought into the form (2).

The form (2), with x_1, y_1 retained, is still obtainable in infinitely many ways. If

$$f = x_1 y_1 v - u^2 = x_1 y_1 v_1 - u_1^2,$$

then

$$x_1 y_1 (v - v_1) = (u - u_1)(u + u_1). \quad (3)$$

Since the intersection of x_1 and y_1 cannot lie on a nonsingular quartic, one of the two factors on the right must be divisible by $x_1 y_1$. After choosing the sign of u_1 , put

$$u_1 = u + \lambda x_1 y_1.$$

Then

$$f = x_1 y_1 (v + 2\lambda u + \lambda^2 x_1 y_1) - (u + \lambda x_1 y_1)^2. \quad (4)$$

If λ can be so chosen that the quadratic form $v + 2\lambda u + \lambda^2 x_1 y_1$ splits into two linear factors, then, with $u + \lambda x_1 y_1 = u_1$,

$$f = x_1 y_1 x_2 y_2 - u_1^2. \quad (5)$$

Thus x_2, y_2 are two further bitangents, and the eight contact points of x_1, y_1, x_2, y_2 lie on the conic $u_1 = 0$.

The necessary and sufficient condition that a ternary quadratic form split into two linear factors is the vanishing of its determinant. If u, v are expressed in the variables x_1, y_1 and a third variable and if their coefficients are denoted by b_{ik} and a_{ik} , one obtains

$$\begin{vmatrix} a_{11} + 2\lambda b_{11} & a_{12} + 2\lambda b_{12} & a_{13} + 2\lambda b_{13} \\ a_{21} + 2\lambda b_{21} & a_{22} + 2\lambda b_{22} & a_{23} + 2\lambda b_{23} + \frac{1}{2}\lambda^2 \\ a_{31} + 2\lambda b_{31} & a_{32} + 2\lambda b_{32} + \frac{1}{2}\lambda^2 & a_{33} + 2\lambda b_{33} \end{vmatrix} = 0. \quad (6)$$

This is an equation of the fifth degree in λ , and so it gives five decompositions

$$x_2y_2, x_3y_3, x_4y_4, x_5y_5, x_6y_6.$$

Thus:

To each pair of bitangents x_1y_1 there belong five further pairs x_iy_i such that the eight points of contact of x_1, y_1, x_i, y_i lie on a conic.

The six pairs

$$x_1y_1, x_2y_2, x_3y_3, x_4y_4, x_5y_5, x_6y_6$$

are called a *Steiner complex*. The word “complex” is used here instead of the older term “Steiner group”, because in algebra the word group has a definite different meaning.

Take three pairs x_1y_1, x_2y_2, x_3y_3 of such a complex. Then

$$f = x_1y_1x_2y_2 - u_1^2 = x_1y_1x_3y_3 - u_2^2,$$

and therefore

$$x_1y_1(x_2y_2 - x_3y_3) = (u_1 - u_2)(u_1 + u_2).$$

As before, it follows that $u_1 - u_2 = hx_1y_1$, and hence

$$4f = 4x_1y_1x_2y_2 - (x_1y_1 + x_2y_2 - x_3y_3)^2. \quad (7)$$

After a harmless change of constant factors this becomes

$$\begin{aligned} -4f &= (x_1y_1 + x_2y_2 - x_3y_3)^2 - 4x_1y_1x_2y_2 \\ &= x_1^2y_1^2 + x_2^2y_2^2 + x_3^2y_3^2 - 2x_2y_2x_3y_3 - 2x_1y_1x_2y_2 - 2x_1y_1x_3y_3. \end{aligned} \quad (8)$$

The equation $f = 0$ may also be written in the elegant irrational form

$$\sqrt{x_1y_1} + \sqrt{x_2y_2} + \sqrt{x_3y_3} = 0. \quad (9)$$

From the symmetry of (9) one obtains:

The pairs of a Steiner complex have the property that the eight points of contact of any two of the pairs lie on a conic, and that one obtains the same complex regardless of which of the six pairs one starts from.

Three bitangents whose contact points lie on a conic are said, after Frobenius, to form a *syzygetic triple*; three bitangents whose six contact points do not lie on a conic form an *azygetic triple*. Similarly four bitangents whose eight contact points lie on a conic form a syzygetic quadruple, and a system of bitangents of which every three are azygetic is called an azygetic system.

Three bitangents occurring in a Steiner complex in such a way that no two of them form one of its pairs, for instance x_1, x_2, x_3 , are always azygetic.

The proof follows from (9). The three lines x_1, x_2, x_3 cannot pass through one point; such a point would be on f and would therefore be singular. Thus we may use x_1, x_2, x_3 as coordinates and write

$$\begin{aligned} y_1 &= \alpha_1x_1 + \alpha_2x_2 + \alpha_3x_3, \\ y_2 &= \beta_1x_1 + \beta_2x_2 + \beta_3x_3, \\ y_3 &= \gamma_1x_1 + \gamma_2x_2 + \gamma_3x_3. \end{aligned}$$

The contact points of x_1, x_2, x_3 are obtained from the three pairs of equations

$$\begin{aligned} x_1 &= 0, & x_2(\beta_2x_2 + \beta_3x_3) - x_3(\gamma_2x_2 + \gamma_3x_3) &= 0, \\ x_2 &= 0, & x_3(\gamma_3x_3 + \gamma_1x_1) - x_1(\alpha_3x_3 + \alpha_1x_1) &= 0, \\ x_3 &= 0, & x_1(\alpha_1x_1 + \alpha_2x_2) - x_2(\beta_1x_1 + \beta_2x_2) &= 0. \end{aligned}$$

If these six points lay on a conic, the coefficients of that conic would have to satisfy the relations

$$a_2 = h_1\beta_2, \quad a_3 = -h_1\gamma_3, \quad a_3 = h_2\gamma_3, \quad a_1 = -h_2\alpha_1, \quad a_1 = h_3\alpha_1, \quad a_2 = -h_3\beta_2, \quad (10)$$

whence $h_1 = -h_2 = -h_3 = -h_1$, impossible unless all h_i vanish. Then the conic would pass through the pairwise intersections of x_1, x_2, x_3 , whereas it was supposed to pass through their contact points. This proves the theorem.